

EDS 223 Assignment 1

Map making practice

Monique Hernandez

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Air Quality and Cancer risk in Oahu, Hawaii via EJScreen Tool

The two maps created focus on the environmental injustice those living on the island of Oahu in Hawaii face when near the areas that demonstrate high risk of cancer from toxic chemicals in the air as listed by the EPA's Toxics Release Inventory (TRI) Program. There is an extra map focusing on the percentage of those living in an area on the island of Oahu who have been determined to be at poverty-level, which was defined as being less than or equal to twice the federal poverty level. This was done to verify the injustice of those in the near vicinity of the areas with the high risk of air cancer.

Map 1 determines the toxic chemical weight, with the EJScreen having raw values range from 0 to 4,512,172.1, with the highest at 28,117.14. Map 2 shows the percentile of high risk areas in danger of cancer from toxic chemicals, which was calculated by persons per million over a 70-year lifetime with the max being 50 of a raw value. Map 3 displays the percentage of those who are considered to be equal to or less than the Federal poverty level.

The burden of those living near the areas with high risk of cancer may impact those who face poverty at an unequal advantage, due to the fact that some may be able to afford air purifiers or the ability to relocate. Lastly, map 1 and map 2 have a strong correlation with the location of two major infrastructures on the island, Daniel K. Inouye International Airport and Par Hawaii Refining, an energy plant, that had a recent settlement in 2022 due to Clean Air Violations with the EPA.

Read EJScreen Data

```
library(tidyverse) # For R
library(sf) # Read vector
library(here) # Locate files
```

```

library(stars)
library(tmap) # For static and interactive maps
library(janitor) # Clean data

# read in geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read(here::here("data",
  "ejscreen", "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"),
  quiet = TRUE)

# filter to a state you are interested in
hawaii <- ejscreen |>
  dplyr::filter(ST_ABBREV == "HI")

# filter to a county you are interested in
honolulu <- ejscreen |>
  dplyr::filter(CNTY_NAME %in% c("Honolulu County"))

# find the average values for all variables within counties
hawaii_counties <- aggregate(hawaii,
  by = list(hawaii$CNTY_NAME),
  FUN = mean)

```

Data Cleaning

```

# Convert columns to lower case

hawaii <- hawaii |>
  clean_names()

honolulu <- honolulu |>
  clean_names() |>
  # Convert decimals to whole numbers
  mutate(honolulu, lipct = lowincpct * 100)

hawaii_counties <- hawaii_counties |>
  clean_names()

# Summarize columns of choice
summary(hawaii$rsei_air)

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NAs
0.00	54.55	177.48	412.46	475.16	28117.14	25

```
summary(hawaii$p_d2_cancer)
```

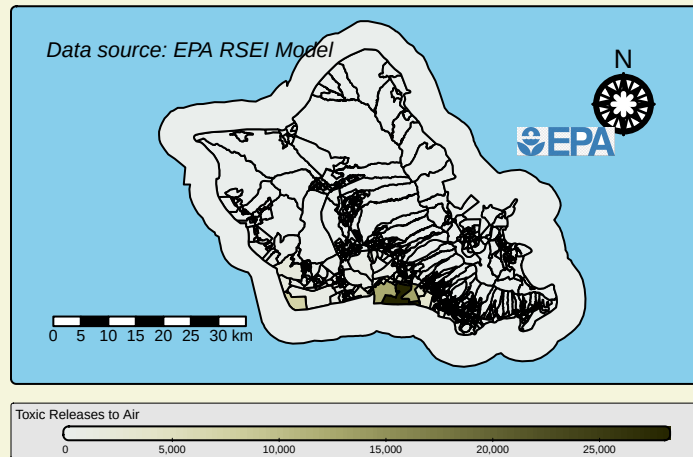
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NAs
0.0	25.0	50.0	49.4	75.0	100.0	30

Map 1: Toxic Air Pollution

```
# Remove missing values
honolulu$rsei_air[is.na(honolulu$rsei_air)] <- 0

tmap_mode("plot")
tm1 <- tm_shape(honolulu,
bbox = c(-158.580780,21.183131,-157.409363,21.7754421)) +
  tm_polygons(fill = "rsei_air",
  fill.legend = tm_legend(
    title = "Toxic Releases to Air",
    orientation = "landscape",
    width = 60),
  fill.scale = tm_scale_continuous(
    values = "scico.broc", midpoint = 30)) +
  tm_compass(
    type = "rose",
    size = 2,
    position = c("right", "top")) +
  tm_scalebar(position = c("left", "bottom")) +
  tm_title(text = "Toxic Air Pollution") +
  tm_layout(style= "natural",
    bg.color = "skyblue",
    frame = TRUE, outer.bg.color = "beige") +
  tm_credits("Data source: EPA RSEI Model",
    fontface = "italic",
    position = c("left", "top")) +
  tm_logo("image/epa.png", height=1,
    position = tm_pos_in(pos.h = 0.7, pos.v = 0.7))
tm1
```

Toxic Air Pollution



Map 2: Percentile of Lifetime Risk of Cancer from Toxic Air

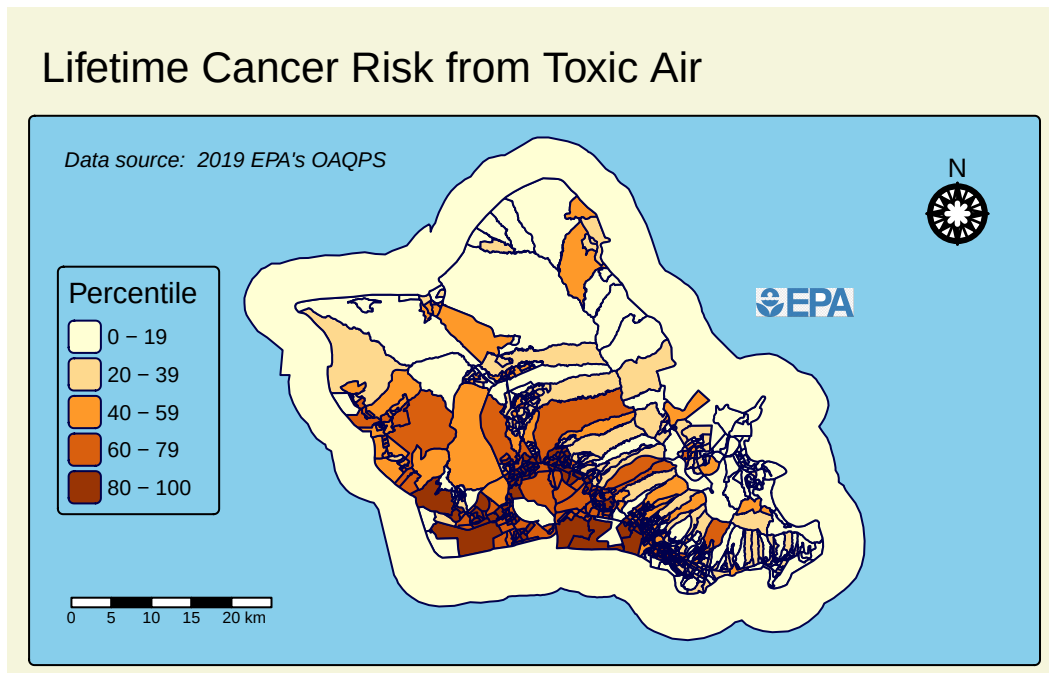
```
# Remove missing values
honolulu$p_d2_cancer[is.na(honolulu$p_d2_cancer)] <- 0

# Select primary spatial object as county of choice
tm2 <- tm_shape(honolulu,
bbox = c(-158.580780,21.183131,-157.409363,21.7754421)) +
  # Map percentile of cancer and create legend
  tm_polygons(fill = "p_d2_cancer",
fill.legend = tm_legend(title = "Percentile",
position = c("left", "center"),
bg.color = "skyblue",
width = 6)) +
  # Customize compass and position
  tm_compass(
    type = "rose",
    size = 2,
    position = c("right", "top")) +
  tm_scalebar(position = c("left", "bottom")) +
  # Add title to map
```

```

tm_title(text = "Lifetime Cancer Risk from Toxic Air") +
# Customize color of backgrounds
tm_layout(style= "beaver",
          bg.color = "skyblue", outer.bg.color = "beige") +
# Add Source
tm_credits("Data source: 2019 EPA's OAQPS",
fontface = "italic", position = c("left", "top")) +
# Add logo
tm_logo("image/epa.png", height=1,
position = tm_pos_in(pos.h = 0.7, pos.v = 0.7))
tm2

```



Map 3: Population Status Determined at Poverty-level

```

tm3 <- tm_shape(honolulu,
bbox = c(-158.580780,21.183131,-157.409363,21.7754421)) +
tm_polygons(fill = "lipct",
fill.legend = tm_legend(title = "Percentage",
position = c("left", "center"),
bg.color = "skyblue",

```

```

width = 6)) +
tm_compass(
type = "rose",
size = 2,
position = c("right", "top")) +
tm_scalebar(position = c("left", "bottom")) +
tm_title(text = "Low Income Population") +
tm_layout(style = "beaver",
          bg.color = "skyblue",
          outer.bg.color = "beige") +
tm_credits("Data source: EjScreen",
fontface = "italic", position = c("left", "top")) +
tm_logo("image/epa.png", height=1,
position = tm_pos_in(pos.h = 0.7, pos.v = 0.7))
# tm_minimap(server = NA, toggle = TRUE, position = c("left", "bottom"))
tm3

```

Low Income Population

